

Module 4 Assignment 4B: Personal Inventory Database

Course: CS 85 - PHP Programming

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For this assignment, I created a PHP page that connects to a MySQL database using PDO and displays personal inventory records from an items table. I created the inventory_db database, added an items table, and inserted more than five sample inventory records.

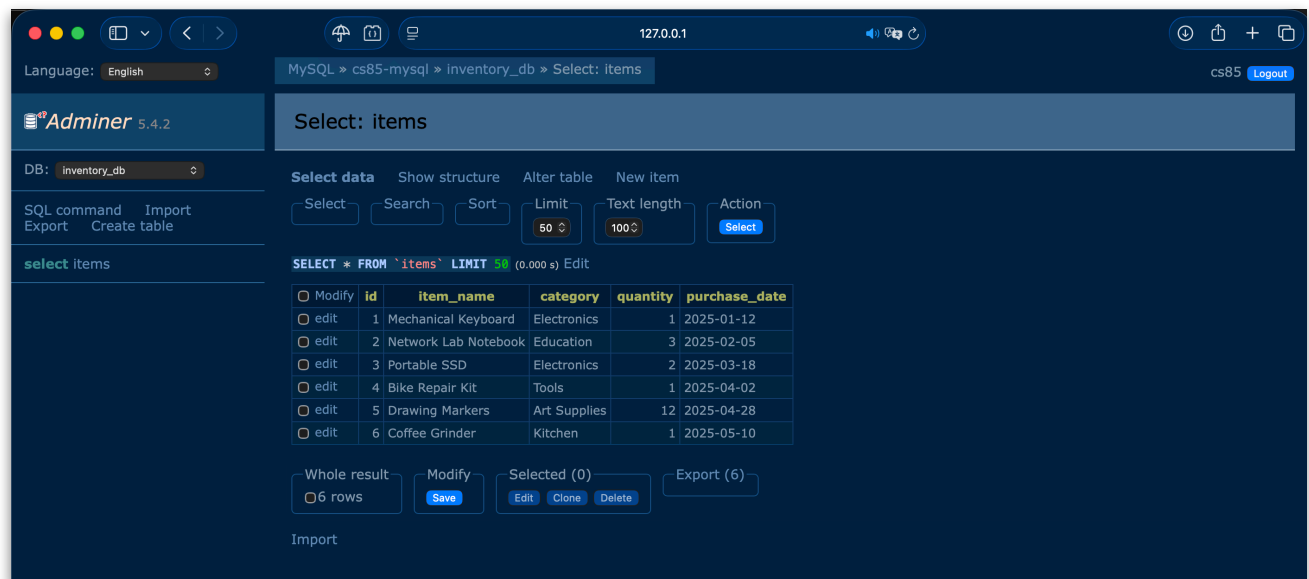
The page uses PDO prepared statements to retrieve inventory data securely. I also escaped output before displaying database values in the HTML table to help protect against XSS. In addition to the basic assignment requirements, I added filtering controls so the inventory can be searched by item/category, filtered by category and minimum quantity, and sorted in different display orders.

Test URL: http://127.0.0.1:8000/module4b/show_inventory.php

GitHub Repository: https://github.com/sergehall/cs85-php-programming/blob/main/public/module4b/show_inventory.php

Screenshots:

Screenshot 1: MySQL database and items table with sample records



The screenshot shows the Adminer 5.4.2 interface for a MySQL database named 'inventory_db'. The 'items' table is selected, and the 'Select data' tab is active. The table contains 6 rows of sample inventory records. The interface includes search, sort, and limit controls, as well as an 'Export' button.

	id	item_name	category	quantity	purchase_date
edit	1	Mechanical Keyboard	Electronics	1	2025-01-12
edit	2	Network Lab Notebook	Education	3	2025-02-05
edit	3	Portable SSD	Electronics	2	2025-03-18
edit	4	Bike Repair Kit	Tools	1	2025-04-02
edit	5	Drawing Markers	Art Supplies	12	2025-04-28
edit	6	Coffee Grinder	Kitchen	1	2025-05-10

Screenshot 2: show_inventory.php code showing PDO connection, prepared statement, and escaped output

```
package.json show_inventory.php X
public > module4b > show_inventory.php

90 if (!isset($allowedSorts[$sortBy])) {
91     $sortBy = 'category_name_asc';
92 }
93
94 try {
95     $dsn = "mysql:host={$databaseHost};port={$databasePort};dbname={$databaseName};charset=utf8mb4";
96     $pdo = new PDO($dsn, $databaseUser, $databasePassword, [
97         PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
98         PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
99         PDO::ATTR_EMULATE_PREPARES => false,
100     ]);
101
102     $categoryStatement = $pdo->query('SELECT DISTINCT category FROM items ORDER BY category ASC');
103     $categories = array_map(static fn (array $row): string => (string) $row['category'], $categoryStatement->fetchAll());
104
105     $where = [];
106     $parameters = [];
107
108     // Filter input values are bound as PDO parameters instead of being mixed into SQL.
109     if ($searchTerm !== '') {
110         $where[] = '(item_name LIKE :search_item OR category LIKE :search_category)';
111         $parameters['search_item'] = '%'.$searchTerm.'%';
112         $parameters['search_category'] = '%'.$searchTerm.'%';
113     }
114
115     if ($selectedCategory !== '') {
116         $where[] = 'category = :category';
117         $parameters['category'] = $selectedCategory;
118     }
119 }
```

```
package.json show_inventory.php X
public > module4b > show_inventory.php

94 try {
127
128     // ORDER BY uses an allowlist because column names cannot be safely bound as values.
129     $statement = $pdo->prepare(
130         "SELECT id, item_name, category, quantity, purchase_date
131         FROM items{$whereSql}
132         ORDER BY {$ordersSql}"
133     );
134     $statement->execute($parameters);
135     $items = $statement->fetchAll();
136     $itemCount = count($items);
137     $connectionMessage = 'Showing '.$itemCount.' inventory ' . ($itemCount === 1 ? 'item' : 'items') . ' from MySQL';
138 } catch (PDOException $exception) {
139     $connectionMessage = 'Could not load inventory records.';
140     $connectionError = $exception->getMessage();
141 }
142 }
```

```
package.json show_inventory.php X
public > module4b > show_inventory.php

141 }
142
143 function h(string $value): string
144 {
145     return htmlspecialchars($value, ENT_QUOTES, 'UTF-8');
146 }
147
148 function formatPurchaseDate(string $value): string
149 {
150     $timestamp = strtotime($value);
151
152     return $timestamp === false ? $value : date('M j, Y', $timestamp);
153 }
```

Screenshot 3: Inventory page displaying records in the browser

The screenshot shows a web browser displaying the 'Personal Inventory Database' application. The page has a dark green header with a 'Back to Module 4' button and the title 'MODULE 4 ASSIGNMENT 4B Personal Inventory Database'. Below the header, a status bar indicates 'Status: Showing 6 inventory items from MySQL.'. The main content area features a search and filter section with four input fields: 'SEARCH ITEM OR CATEGORY' (containing 'keyboard, tools, kitchen...'), 'CATEGORY' (a dropdown menu set to 'All categories'), 'MINIMUM QUANTITY' (a numeric input field), and 'DISPLAY ORDER' (a dropdown menu set to 'Category, then item A-Z'). Below these fields are 'Apply Filters' and 'Reset' buttons. The 'Inventory Items' section displays a table with the following data:

ITEM	CATEGORY	QUANTITY	PURCHASE DATE
Drawing Markers	Art Supplies	12	Apr 26, 2025
Network Lab Notebook	Education	3	Feb 5, 2025
Mechanical Keyboard	Electronics	1	Jan 12, 2025
Portable SSD	Electronics	2	Mar 18, 2025
Coffee Grinder	Kitchen	1	May 10, 2025
Bike Repair Kit	Tools	1	Apr 2, 2025

Below the table, the 'Database Setup' section contains a yellow box with text: 'The database is named `inventory_db`. The inventory table is named `items` and contains at least five personal inventory records. The full SQL setup script is included in the PHP comments at the top of this file.' The 'Reflection' section follows with a paragraph: 'I chose practical items from a student workspace and home lab because they make the inventory feel realistic. This could scale into a real system by adding users, locations, suppliers, low-stock alerts, and audit history. PDO prepared statements help protect against SQL injection.'

Screenshot 4: Inventory page using filters/search/sort controls

This screenshot shows the same 'Personal Inventory Database' application, but with filters applied. The status bar now indicates 'Status: Showing 2 inventory items from MySQL.'. The 'SEARCH ITEM OR CATEGORY' field still contains 'keyboard, tools, kitchen...', but the 'CATEGORY' dropdown is now set to 'Electronics'. The 'MINIMUM QUANTITY' and 'DISPLAY ORDER' fields remain unchanged. The 'Inventory Items' table now displays only two items:

ITEM	CATEGORY	QUANTITY	PURCHASE DATE
Mechanical Keyboard	Electronics	1	Jan 12, 2025
Portable SSD	Electronics	2	Mar 18, 2025